CrowdWisdomTrading: Quantitative Prediction Report

Multi-Horizon Sentiment & Market Price Evaluation | Generated 2026-10-19 16:31 UTC

1. Executive Summary & Free-First Architecture

This report documents out-of-sample predictive performance across 6 horizons (n+0 to n+5) using exclusively free tools, local VADER NLP sentiment analysis, and chronological purged walk-forward cross validation without lookahead leakage.

2. Walk-Forward Validation Metrics (Out-of-Sample)

Horizon	Model	MAE ($)	RMSE ($)	MAPE (%)	Dir Acc (%)	Base MAE ($)
n+0	Persistence	$187.0	$291.5	0.3%	49.7%	$187.0
n+0	Ridge	$187.4	$292.3	0.3%	51.5%	$187.0
n+0	Random Forest	$187.4	$292.3	0.3%	51.1%	$187.0
n+0	Hist Gradient	$187.8	$293.0	0.3%	51.5%	$187.0
n+1	Persistence	$265.5	$408.6	0.4%	50.2%	$265.5
n+1	Random Forest	$266.8	$410.9	0.4%	51.1%	$265.5
n+1	Ridge	$267.1	$410.1	0.4%	51.8%	$265.5
n+1	Hist Gradient	$267.1	$411.1	0.4%	50.8%	$265.5
n+2	Persistence	$326.9	$494.7	0.5%	50.0%	$326.9
n+2	Random Forest	$327.2	$497.8	0.5%	52.9%	$326.9
n+2	Hist Gradient	$328.4	$498.4	0.5%	51.8%	$326.9
n+2	Ridge	$329.2	$497.6	0.5%	51.3%	$326.9
n+3	Persistence	$375.9	$564.2	0.5%	50.7%	$375.9
n+3	Random Forest	$378.0	$568.8	0.5%	51.7%	$375.9
n+3	Hist Gradient	$380.8	$571.8	0.5%	50.7%	$375.9
n+3	Ridge	$380.8	$569.9	0.5%	51.2%	$375.9
n+4	Persistence	$421.9	$623.7	0.6%	50.6%	$421.9
n+4	Random Forest	$426.9	$631.8	0.6%	50.2%	$421.9

3. Quantitative Trading Strategy Performance

Model	Horizon	Cum Ret (%)	Exposure (%)	Sharpe	Sortino	Max DD (%)	Trades
No Trade Basel	n+1	+0.0%	0.0%	0.00	0.00	0.0%	0
Ridge	n+1	-8.8%	4.2%	-1.88	-2.73	-12.6%	154
Random Forest	n+1	+0.2%	4.3%	-0.21	-0.35	-12.4%	112
Hist Gradient	n+1	-9.8%	3.3%	-2.60	-3.62	-13.9%	98